

Assesement 1

CompTIA A+ Troubleshooting & Helpdesk



SECTION A – CONCEPT APPLICATION

SCENARIO 1

You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question:

Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

To select compatible hardware for a new office workstation, I would follow these steps:

1. **Choose the CPU first**
 - Check the CPU's **socket type** and generation.
 - Example: If the CPU uses an **AM5 socket**, the motherboard must support AM5.
2. **Choose a compatible motherboard**
 - The motherboard must support the CPU's **socket and generation**.
 - Check the motherboard's supported **RAM type**, such as DDR4 or DDR5.
 - Also check the required storage connections, such as **SATA or M.2**.
3. **Choose the RAM**
 - Select the RAM type supported by the motherboard.
 - For example, if the motherboard supports **DDR5**, use DDR5 RAM.
 - Check the maximum RAM capacity and supported speed.
4. **Choose the storage device**
 - Check whether the motherboard supports the selected storage.
 - For an **M.2 SSD**, make sure the motherboard has a compatible M.2 slot.
 - For a **SATA SSD/HDD**, make sure SATA ports are available.
5. **Confirm compatibility**
 - Check the **CPU and motherboard manufacturer's specifications**.
 - Check the motherboard's **CPU Support List (QVL)**.
 - Check RAM compatibility and motherboard specifications.
 - Make sure all required slots and connectors are available.

SCENARIO 2

You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question:

Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Storage	Speed	Cost	Main Feature
HDD	Slow	Cheap	Uses moving parts
SSD	Fast	Medium	No moving parts
NVMe SSD	Very Fast	Higher	Uses PCIe interface

1. HDD

- HDD uses a **spinning disk and moving parts**.
- It is slower than SSD.
- It is cheaper and available in large capacities.
- Good for storing large amounts of data at low cost.

2. SSD

- SSD has **no moving parts**.
- It is much faster than HDD.
- It makes Windows and applications start faster.
- It is also quieter and uses less power.

3. NVMe SSD

- NVMe is a type of SSD that uses the **PCIe interface**.
- It is faster than a normal SATA SSD.
- It provides very fast data transfer and low latency.
- It is usually connected through an **M.2 slot**.

Recommendation

For office employees using applications such as **Microsoft Word, Excel, PowerPoint, email, and web browsers**, I would recommend an **SSD, preferably an NVMe SSD**.

Technical reason: NVMe provides faster read/write speeds and lower latency, so the computer can **boot faster, open applications quickly, and load files faster**. If the budget is limited, a normal **SATA SSD** is also a very good choice for office work.

SCENARIO 3

You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question:

Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

For a workstation running **graphics-intensive design software for several hours**, the **GPU and cooling system** are very important.

1. Role of the GPU

- The GPU processes **graphics and visual data**.
- It helps design software run smoothly.
- A good GPU improves **rendering, 3D work, and screen performance**.
- It reduces the workload on the CPU.

2. Role of the Cooling System

- When the GPU and CPU work for many hours, they produce **heat**.
- The cooling system uses **fans, heatsinks, and sometimes liquid cooling** to remove this heat.
- Good cooling prevents the components from becoming too hot.
- It helps prevent **thermal throttling**, where the system slows down because of high temperature.
- It also improves system stability and helps protect the hardware.

SCENARIO 4

You are assembling a PC from components sourced from three different vendors.

Question:

Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Hardware compatibility means making sure that all computer components can work together properly.

Why is compatibility important?

- The **CPU must match the motherboard socket**.
- The **RAM type** must be supported by the motherboard, such as DDR4 or DDR5.
- The motherboard must have the correct **slots and connectors** for storage devices.
- The **PSU** must provide enough power for all components.
- The **GPU** must fit the motherboard and PC case and have the required power connection.

Problems caused by incompatible components

If components are incompatible, problems can include:

- The computer may **not start or boot**.
- RAM may **not work**.
- The CPU may not be supported by the motherboard.
- Storage devices may not be detected.
- The system may **crash or become unstable**.
- Components may **overheat**.
- In some cases, incorrect power connections can **damage hardware**.

SCENARIO 5

You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question:

Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

Before powering on the PC, I would choose the correct **PSU (Power Supply Unit)** and check all connections carefully.

1. Choosing the PSU

- Calculate the **total power required** by the CPU, GPU, motherboard, storage, and other components.
- Choose a PSU with **more wattage than the estimated requirement** to provide some safety margin.
- Check that the PSU has the required **80 PLUS efficiency rating** and is from a reliable manufacturer.
- Make sure it has the correct connectors, such as **24-pin motherboard, CPU 4/8-pin, GPU PCIe, and SATA power connectors**.

2. Safety Checks

- Make sure the PC is **switched off and unplugged** from the wall.
- Check that there are **no loose screws or metal objects** inside the case.
- Make sure all components are properly installed.
- Do not force any connector into the wrong socket.
- Keep cables away from fans.

3. Connection Checks

- Connect the **24-pin ATX cable** to the motherboard.
- Connect the **CPU power cable** to the motherboard.
- Connect the required **PCIe power cables** to the GPU.
- Connect **SATA power cables** to SATA SSDs/HDDs.
- Check that every connector is fully inserted and secure.

SCENARIO 6

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question:

Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

POST (Power-On Self-Test) is a check performed by the computer when it starts. It checks important hardware such as the **CPU, RAM, motherboard, and storage**.

Checks for successful POST

1. Press the **power button** and check that the system starts normally.
2. Check that the **fans are spinning** and there are no unusual sounds.
3. Check the monitor for the **BIOS/UEFI screen** or manufacturer logo.
4. Make sure the **RAM and CPU are detected** in BIOS/UEFI.
5. Check that the **SSD/HDD is detected**.
6. Check for any **beep codes or error messages**.
7. Confirm that the system continues to the **operating system** without errors.

If POST fails

- Turn off and unplug the computer.
- Check that the **RAM, CPU, GPU, and power cables** are properly connected.
- Reseat the RAM and try starting the system again.

- Check the motherboard's **beep codes or diagnostic LEDs** for the faulty component.
- Disconnect unnecessary devices and test again.
- If needed, reset the **BIOS/UEFI settings (CMOS)**.
- Replace or test the suspected faulty component.

SECTION B — PRACTICAL CODING TASKS

Task 1: Component Identification & Functions

- ❖ Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.

Computer Components and Their Functions —

1. **Motherboard**
 - It is the **main circuit board** of the computer.
 - It connects all components like CPU, RAM, GPU, and storage.
 - It allows the components to communicate with each other.
2. **CPU (Central Processing Unit)**
 - It is the **brain of the computer**.
 - It processes instructions and performs calculations.
 - It runs programs and applications.
3. **RAM (Random Access Memory)**
 - RAM is the computer's **temporary working memory**.
 - It stores data and programs that are currently being used.
 - More RAM helps the computer run multiple programs smoothly.
4. **GPU (Graphics Processing Unit)**
 - It processes **images, videos, and graphics**.
 - It is important for gaming, video editing, and 3D design.
5. **Storage Device (HDD/SSD)**
 - It stores data **permanently**.
 - It stores the operating system, applications, documents, photos, and videos.
 - SSDs are generally faster than HDDs.
6. **Cooling System**
 - It removes **heat** from the CPU and other components.
 - It usually uses fans and heatsinks.
 - It prevents overheating and helps the computer work safely.

In short:

Motherboard = connects everything

CPU = processes data

RAM = temporary memory

GPU = graphics

Storage = saves data

Cooling = removes heat

❖ **Note the make/model or key specification of each component where visible**

For each computer component, note down its **make (brand), model, or important specification** if it is visible.

Component	What to Note
Motherboard	Brand, model, socket type
CPU	Brand, model, generation, GHz
RAM	Brand, capacity (GB), DDR4/DDR5, speed
GPU	Brand, model, VRAM (GB)
Storage	Brand, model, capacity, HDD/SSD/NVMe
Cooling System	Type, brand/model if visible

Example:

- Motherboard: ASUS PRIME B550M-A
- CPU: AMD Ryzen 5 5600
- RAM: 16 GB DDR4
- GPU: NVIDIA RTX 3060 12 GB
- Storage: 1 TB NVMe SSD
- Cooling: CPU Air Cooler

In short: Check each component and write down its **brand, model, and main specification** wherever it is visible.

❖ Present the findings as a labeled component list or table.

Component	Make / Model	Key Specification
Motherboard	ASUS PRIME B550M-A	AM4 Socket
CPU	AMD Ryzen 5 5600	6 Cores, 12 Threads
RAM	16 GB DDR4	3200 MHz
GPU	NVIDIA GeForce RTX 3060	12 GB VRAM
Storage	1 TB NVMe SSD	PCIe NVMe
Cooling System	Air Cooler	Fan + Heatsink

TASK 2: MOTHERBOARD PORTS & SOCKET IDENTIFICATION

❖ Identify and label all major motherboard ports (SATA, power, USB headers, etc.).

Major Motherboard Ports – Easy Answer

The major motherboard ports and connectors are

Port / Connector	Function
24-pin ATX Power	Supplies power to the motherboard
4/8-pin CPU Power	Supplies power to the CPU
SATA Ports	Connect HDDs and SATA SSDs
M.2 Slot	Connects an M.2/NVMe SSD
RAM Slots (DIMM)	Used to install RAM
PCIe Slot	Used for GPU and other expansion cards
USB Headers	Connect front-panel USB ports
Front Panel Header	Connects Power/Reset buttons and LEDs

Fan Headers	Connect CPU and case fans
Audio Header	Connects front-panel audio
24-pin/CPU power connectors	Provide power to motherboard and CPU
Rear USB Ports	Connect external USB devices
LAN/Ethernet Port	Connects the computer to a network
Audio Ports	Connect speakers, headphones, or microphone
HDMI/DisplayPort	Connects a monitor when supported by the motherboard/CPU

❖ Identify all expansion slots (PCIe, M.2, etc.) and their purpose.

Expansion Slots – Easy Answer

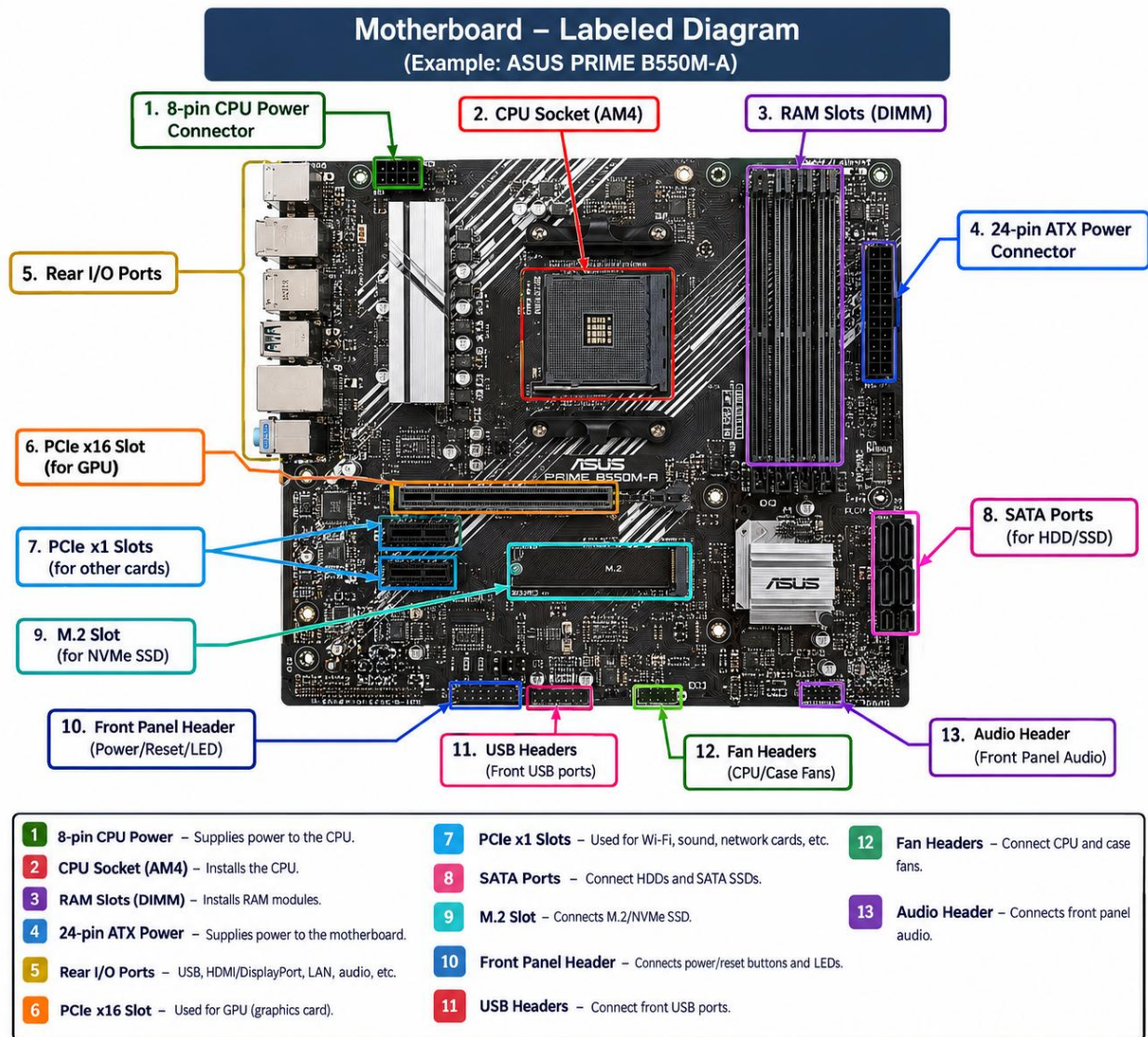
Expansion slots are used to **add extra hardware or features** to a computer.

Expansion Slot	Purpose
PCIe x16	Used mainly for a GPU/graphics card
PCIe x4 / x8	Used for network cards, storage cards, capture cards, etc.
PCIe x1	Used for smaller expansion cards like Wi-Fi, sound, or network cards
M.2 Slot	Used mainly for NVMe SSDs ; some M.2 slots also support Wi-Fi cards
RAM/DIMM Slots	Used to install RAM (technically memory slots, not expansion slots)
PCI (older)	Used for older expansion cards; uncommon in modern PCs

In short:

- PCIe x16 → GPU
- PCIe x1 → Wi-Fi/Sound/Network cards
- M.2 → NVMe SSD
- RAM slots → RAM

- ❖ Photograph or diagram the motherboard with labels.



Task 3: RAM & Storage Installation

- ❖ Safely remove the RAM module, noting ESD precautions taken.

Safe RAM Removal – Easy Steps

1. **Shut down the computer** and unplug the power cable.
2. Press the **power button for a few seconds** to discharge remaining power.
3. Wear an **anti-static wrist strap** or touch a grounded metal part of the case to prevent ESD.

4. Do not touch the **gold contacts** on the RAM module.
5. Open the **locking clips** on both sides of the RAM slot.
6. Hold the RAM by its **edges** and gently pull it straight out.
7. Place the RAM on an **anti-static bag or surface**.

ESD precautions: I would use an anti-static wrist strap, handle the RAM only by its edges, and avoid touching its contacts to prevent electrostatic damage.

❖ **Reinstall the RAM module correctly and confirm it is seated properly.**

Reinstall RAM –

1. **Turn off the computer** and unplug the power cable.
2. Wear an **anti-static wrist strap** or touch the metal case to prevent ESD.
3. Check the **RAM slot and RAM notch**. The notch must match the slot.
4. Hold the RAM by its **edges** and place it into the correct slot.
5. Press the RAM down **evenly on both ends** until it clicks into place.
6. Check that the **locking clips close automatically** and hold the RAM firmly.
7. Make sure the RAM is **straight and fully inserted**.
8. Start the computer and check the **BIOS/UEFI** to confirm the correct RAM capacity is detected.

Confirmation: The RAM is properly seated when the clips are locked, the module is level, and the BIOS/UEFI detects the expected amount of RAM.

- ❖ **Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).**

Remove and Reinstall Storage Device – Easy Steps

1. **Shut down the computer** and unplug the power cable.
2. Take **ESD precautions** by using an anti-static wrist strap or touching the metal case.
3. **Identify the storage type:**
 - **SATA:** Uses a SATA data cable and SATA power cable.
 - **NVMe:** Usually fits directly into an **M.2 slot** on the motherboard.
4. **Remove the storage:**
 - For SATA, disconnect the **SATA data and power cables**.
 - For NVMe, remove the **M.2 screw**, then gently remove the drive.
5. **Reinstall the storage:**
 - For SATA, reconnect the SATA data and power cables firmly.
 - For NVMe, insert the drive into the M.2 slot at an angle, press it down, and secure it with the screw.
6. Start the computer and check **BIOS/UEFI** to confirm the storage device is detected.

Connection type used: Write down whether the device is **SATA or NVMe (M.2)**.

Confirmation: The storage device is correctly installed when it is firmly connected and **appears in BIOS/UEFI**.

- ❖ **Confirm both components are detected correctly after reinstallation.**

Confirm Components Are Detected –

After reinstalling the **RAM and storage device**, I would check that both are working correctly.

1. **Start the computer** and enter the **BIOS/UEFI**.
2. Check the **RAM information:**
 - Confirm the correct RAM capacity is shown, such as **8 GB or 16 GB**.
3. Check the **storage information:**
 - Confirm the **HDD/SSD/NVMe** is listed.
 - Check that its capacity is correct.
4. Start Windows and open **Task Manager** → **Performance** to check the RAM.
5. Open **Disk Management** or **This PC** to confirm the storage is detected.

Confirmation: If the correct **RAM capacity** and **storage device/capacity** are shown without errors, both components have been reinstalled correctly.

TASK 4: FULL DESKTOP PC ASSEMBLY

- ❖ **Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU**

PC Assembly – Easy Steps

- 1. Prepare the workspace**
 - Turn off and unplug everything.
 - Use an **anti-static wrist strap** to prevent ESD damage.
- 2. Install the CPU**
 - Open the CPU socket on the motherboard.
 - Align the CPU correctly with the socket mark.
 - Gently place it and lock the socket.
- 3. Install the RAM**
 - Open the RAM slot clips.
 - Match the RAM notch with the slot.
 - Press down until the clips lock.
- 4. Install the storage**
 - For **NVMe**, insert it into the M.2 slot and secure it with the screw.
 - For **SATA**, connect the SATA data cable and SATA power cable.
- 5. Install the motherboard**
 - Place the motherboard into the case.
 - Align it with the case holes and secure it with screws.
- 6. Install the GPU**
 - Insert the GPU into the correct **PCIe x16 slot**.
 - Secure it to the case with a screw.
 - Connect the required GPU power cable from the PSU.
- 7. Install the PSU**
 - Secure the PSU inside the case.
 - Connect the **24-pin motherboard power** and **CPU power** cables.
 - Connect power cables to the storage and GPU as required.
- 8. Final check**
 - Check that all components and cables are properly connected.

- Make sure no cables are touching the fans.
 - Connect the monitor, keyboard, and power cable.
9. **First boot**
- Switch on the PSU and press the PC power button.
 - Check that the system starts and enters **BIOS/UEFI**.

❖ **Follow correct installation sequence and ESD/safety precautions throughout.**

Installation Sequence and Safety –

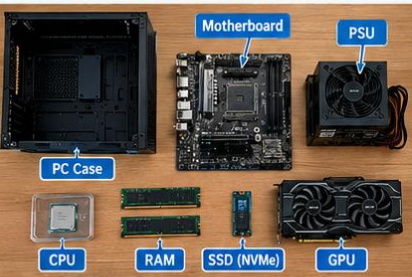
I would follow the correct installation order and take **ESD (Electrostatic Discharge)** precautions throughout.

1. **Prepare the workspace**
 - Switch off and unplug the power supply.
 - Keep the work area clean and dry.
 - Use an **anti-static wrist strap**.
2. **Install CPU**
 - Carefully place the CPU in the correct socket.
 - Do not touch the CPU pins or contacts.

3. **Install RAM**
 - Match the RAM notch with the slot.
 - Press it down until the clips lock.
4. **Install Storage**
 - Install the **NVMe SSD in the M.2 slot** or connect the **SATA drive** with its cables.
5. **Install Motherboard**
 - Place it correctly inside the case.
 - Secure it with the correct screws.
6. **Install GPU**
 - Insert the GPU into the correct PCIe slot.
 - Secure it properly and connect its power cable if required.
7. **Install PSU and connect cables**
 - Connect the **24-pin motherboard cable** and **CPU power cable**.
 - Connect GPU and storage power cables as required.
8. **Final safety check**
 - Make sure all components are firmly installed.
 - Check that no loose screws are inside the case.
 - Keep cables away from fans.
 - Check all connections before powering on.

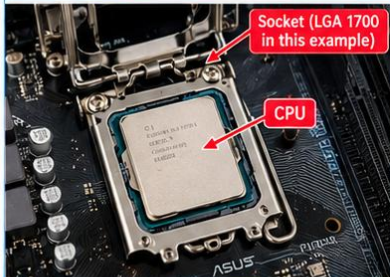
❖ **Document each step with labeled photos or diagrams.**

1. Preparation (Workspace & Components)



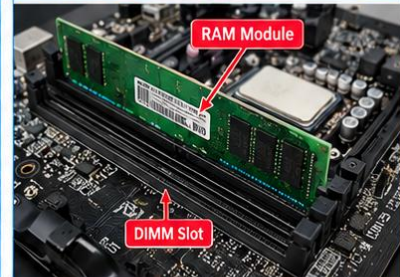
- ✓ **Safety / ESD Precautions:**
- Power off and unplug everything.
 - Use an anti-static wrist strap.
 - Keep the workspace clean and dry.

2. CPU Installation



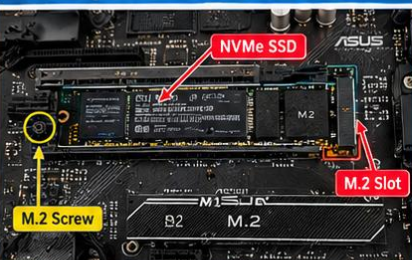
1. Open the CPU socket lever.
2. Align the CPU with the triangle mark.
3. Gently place the CPU (do not touch the pins).
4. Close the lever and lock it.

3. RAM Installation



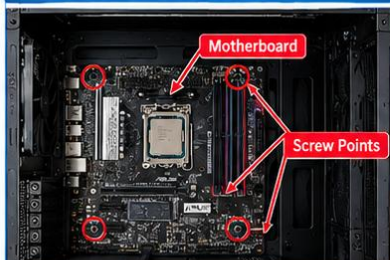
1. Open the slot clips.
2. Check the notch in the RAM and match it with the slot.
3. Insert the RAM and press down until the clips lock.

4. Storage Installation (NVMe SSD)



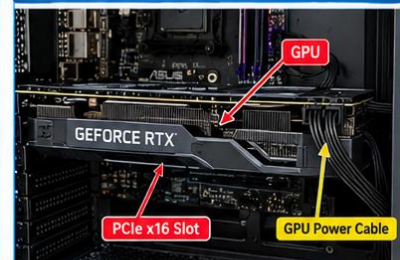
1. Insert the NVMe SSD into the M.2 slot at an angle.
 2. Press it down and secure with the screw.
- (For SATA, connect data and power cables.)

5. Motherboard Installation



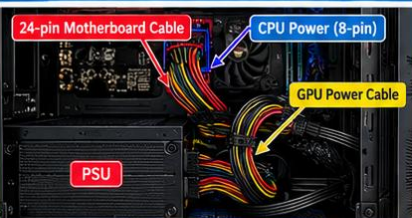
1. Place the motherboard in the case.
2. Align with the mounting holes.
3. Secure with screws (use standoffs if required).

6. GPU Installation



1. Insert the GPU into the PCIe x16 slot.
2. Secure it with a screw to the case.
3. Connect the GPU power cable from the PSU.

7. PSU Installation and Cable Connections



1. Install the PSU in the case.
2. Connect 24-pin motherboard cable.
3. Connect 8-pin CPU power cable.
4. Connect GPU and storage power cables (as required).

8. Final Check (Fully Assembled PC)



- ✓ All components are properly installed.
- ✓ Cables are connected firmly.
- ✓ No loose screws inside the case.
- ✓ Cables are not touching fans.
- ✓ Monitor, keyboard and power cable are connected.



Build is ready for first boot!

9. First Boot – BIOS/UEFI



- Check that the CPU, RAM and storage are detected in BIOS/UEFI.
- Then install the OS and drivers.

❖ **Power on the system and confirm it passes POST successfully.**

Power On and Check POST – Easy Answer

1. Connect the **monitor, keyboard, mouse, and power cable**.
2. Switch on the **PSU** and press the computer's **Power button**.
3. Check that the **fans start spinning** and the monitor displays the startup screen.
4. Watch for the **BIOS/UEFI screen or manufacturer logo**.
5. Check that the **CPU, RAM, and storage** are detected.
6. Make sure there are **no error messages or beep codes**.
7. Confirm that the computer continues to **boot into the operating system**.

Confirmation: The system has passed POST successfully if it starts normally, shows no POST errors, detects the main hardware, and continues to boot into the operating system.

SECTION C —

MINI CAPSTONE PROJECT

Mini Project: Workstation Deployment & Boot-Failure Diagnosis Objective:

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

- ❖ Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why

Three Hardware Components to Check First

If a newly assembled PC **does not boot**, I would check these three components first:

1. **PSU (Power Supply Unit)**
 - Check that the PSU is switched on and connected correctly.
 - Check the **24-pin motherboard power** and **CPU power cable**.
 - If there is no power, the PC may not start at all.
 2. **RAM**
 - Check that the RAM is **properly seated** in the slot.
 - Remove and reinstall it if necessary.
 - Loose or incompatible RAM can cause **no display or POST errors**.
 3. **CPU**
 - Check that the CPU is **correctly installed** in the motherboard socket.
 - Make sure the CPU power cable is connected.
 - Also confirm that the CPU is **compatible with the motherboard and BIOS**.
-
- ❖ Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

Motherboard–CPU Compatibility – Easy Answer

Motherboard–CPU compatibility is important because the **CPU must match the motherboard's socket and supported CPU list**. If they are not compatible, the computer may not boot or complete POST.

Example of the system inspected:

- **Motherboard:** ASUS PRIME B550M-A
- **CPU:** AMD Ryzen 5 5600
- **Socket:** AM4

The motherboard has an **AM4 socket**, and the AMD Ryzen 5 5600 is an **AM4 CPU**, so they are compatible. The motherboard's CPU support list and required BIOS version should also be checked.

If I used an incompatible CPU, the system could show **no display, fail POST, or not detect the CPU**.

- ❖ **Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.**

RAM and Storage Verification – Easy Answer

After installing the **RAM and storage device**, I would check them as follows:

1. Check RAM

- Make sure the RAM is fully inserted and the **locking clips are closed**.
- Turn on the computer and enter **BIOS/UEFI**.
- Check that the correct **RAM capacity** is shown, for example, **16 GB**.
- During POST, there should be **no RAM error or beep code**.

2. Check Storage

- Make sure the **SATA cables** are connected properly for a SATA drive, or the **M.2 SSD** is secured correctly for NVMe.
- Enter **BIOS/UEFI** and check the storage section.
- Confirm that the correct **SSD/HDD/NVMe drive** and its capacity are detected.
- During POST, there should be **no storage detection error**.

- ❖ Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.

- ❖ **Hardware Installation Checklist**

- ❖ To prevent hardware failures in future deployments, I would use this checklist:

Step	Check
1. Check Compatibility	Confirm CPU, motherboard, RAM, GPU, storage, and PSU are compatible before installation.
2. Follow ESD Safety	Use an anti-static wrist strap and handle components carefully to prevent ESD damage.
3. Check Connections	Make sure RAM, CPU, GPU, storage, and all power cables are properly and firmly connected.
4. Check BIOS/UEFI	After assembly, confirm that CPU, RAM, and storage are detected correctly.
5. Perform POST Test	Power on the PC and check for error messages, beep codes, or hardware detection problems.
6. Document the Build	Record component models, specifications, and installation checks for future troubleshooting.

- ❖ Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

Technical Justification for Using SSD Instead of HDD

For the **ABC Solutions computer lab**, I would use **SSDs instead of HDDs** in modern office workstations.

- **Faster speed:** SSDs have much faster read/write speeds, so Windows and office applications open quickly.
- **Faster boot time:** The computer starts faster compared with an HDD.
- **Better performance:** Employees can work smoothly with applications such as **Word, Excel, PowerPoint, email, and web browsers**.
- **No moving parts:** SSDs have no spinning disks, so they are more resistant to physical shocks and have less mechanical failure risk.
- **Less noise and power:** SSDs operate quietly and generally use less power than HDDs.
- **Better for daily office use:** Since the lab's computers will be used regularly for productivity work, faster loading and response times improve the overall user experience.

SECTION D – AI-AUGMENTED LEARNING

STEP 1 · BUILD WITH AI

► Exact Prompt Given to the AI:

Create a simple step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Arrange the checks in the correct priority order. Cover the motherboard, RAM, storage, and PSU as possible failure points. Keep the checklist easy to follow on-site and apply it to the ABC Solutions scenario, where some newly assembled PCs fail to start or show hardware detection errors.

+ AI's Original Checklist:

Æ Check whether the PC powers on and look for error messages.

Æ Check the motherboard for visible damage or loose connections.

Æ Check whether the RAM is installed correctly and reseal it if necessary.

Æ Check whether the storage device is connected and detected.

24/28

Æ Check the PSU cables and power connections.

Æ Enter BIOS/UEFI and check whether the components are detected.

Æ Try starting the operating system again.

+ Application to the ABC Solutions Scenario:

In the ABC Solutions lab, the checklist would help identify an incorrectly seated RAM module causing a hardware detection or boot problem. Step 3 would catch this issue by checking and reseating the RAM before continuing with further diagnosis.

Step 1 - Test & Debug (Without AI).

► Issue Found in the AI Checklist:

The original checklist does not mention ESD precautions before handling internal components. It also starts checking components without clearly stating that the system should be powered off and disconnected from power first.

+ Corrected Checklist:

Æ Power off and unplug the PC and follow proper ESD precautions before opening the case.

Æ Check the PSU and confirm that the 24-pin motherboard, CPU, and GPU power connectors are properly connected.

Æ Check the RAM and make sure it is fully seated with the retaining clips locked.

Æ Check the motherboard for loose connections, incorrect installation, or diagnostic indicators.

Æ Check the storage device and confirm that the SATA or M.2 connection is secure.

Æ Power on the system and check POST/BIOS/UEFI for hardware detection or error messages.

Æ If the hardware is detected correctly, continue with the operating system boot check.

+ What I Changed and Why:

I added ESD precautions and power disconnection as the first step because internal components should not be handled while the system is powered. I also placed the physical hardware checks before software-level checks. This makes the troubleshooting process safer and more logical for a junior technician.